



Three charts explain US "race" to dominate AI

"Race to achieve global dominance" is a zero-sum game

Almost three years since the launch of ChatGPT, fears of AI robots taking over the world in the decades to come have been overtaken by fears of rivals using it to do so much sooner.

The US will do "whatever it takes to lead the world in AI," President Donald Trump said yesterday at an event to launch a 23-page White House AI Action Plan reinforced by three executive orders.

America is in a "race to achieve global dominance in AI," the paper says. It presents AI as "an industrial revolution, an information revolution, and a renaissance – all at once", but also as a zero-sum game that may "reshape the global balance of power".

The US needs to dismantle regulation that might hinder innovation, build out its infrastructure and energy capacity, and ensure that allies use cutting-edge American technology while rivals – namely China – face tighter export controls, according to the plan.

One of Trump's first acts on taking office in January was to tear up the Biden Administration's Executive Order on AI from October 2023, with its title underling its focus on the "[Safe, Secure and Trustworthy Development and Use of AI](#)", and to issue his own, entitled "[Removing Barriers to American Leadership in AI](#)".

Three charts explain why this is such a priority for this administration:

1. Innovation: power to companies

The US is clearly the world's leading force in AI. It plays host to 42 of the top 50 AI companies, 33 of them in California, according to Forbes. US companies pocketed almost three quarters of the private equity and venture capital investments last year, data from the Stanford AI Index show. (The minimum investment in the study was \$1.5m.) Forty of the most notable AI models last year came from the US, compared with 15 from China and just three from Europe.

The US has not taken the aggressive regulatory approach of the European Union, whose comprehensive AI Act is currently rolling out amid an ongoing debate about how it should be implemented.

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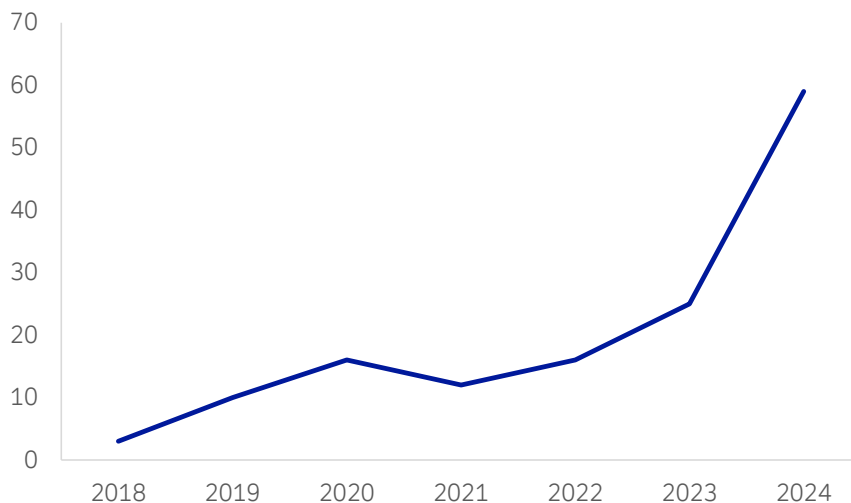
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Instead, it has mainly relied on applying existing laws that already cover aspects such as health and safety, copyright and data privacy. California's state legislature approved what would have been among the most sweeping rules at a state level but Governor Gavin Newsom then vetoed them.

Nonetheless, AI-related regulations introduced by US federal agencies doubled last year to 59, and were issued by 42 agencies, twice as many as the year before. The Health and Human Services Department was particularly prolific, introducing 14.

Figure 1: AI-related regulations doubled from 2023 to 2024



Source : Stanford University HAI AI Index Report 2025, Deutsche Bank

The tide has turned, with Silicon Valley leaders taking high-profile roles in the administration, including two of the three signatories to the Action Plan: Michael Kratsios, White House Office of Science and Technology Policy Director, and a former venture capitalist and protégé of PayPal founder Peter Thiel; and David Sacks, AI and crypto czar, and a leading technology investor now known as one of the hosts of the All-In podcast.

There has also been a surge in lobbying on AI, with more than 500 companies seeking to influence the White House and Congress between January and June, the Financial Times reported yesterday, citing new federal disclosures.

The Action Plan puts cutting red tape front and centre in accelerating AI innovation, saying the federal government should withhold funding from states with burdensome AI regulations and urging several agencies to evaluate and repeal policies that impede innovation.

It also recommends that the Federal Trade Commission review investigations and orders to ensure that they do not "unduly burden AI innovation". That could ease the pressure on companies such as Microsoft, which is subject to a lengthy antitrust probe. President Trump said yesterday that he had contemplated breaking up Nvidia, the \$4trn AI chip producer, before discovering that would be "very hard".



Other measures include encouraging open-source AI; investing in AI-enabled science, despite the administration's recent cuts to university grants; and fostering more widespread adoption, including in government.

The plan was broadly welcomed by the tech industry, prompting some critics to brand it a sell-out, but there is a perhaps surprising underlying emphasis on risk management throughout the plan, ranging from interpretability to cybersecurity.

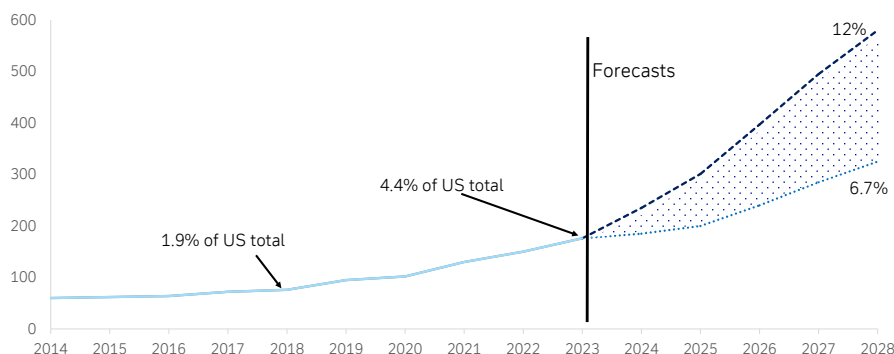
2. Infrastructure: powering up

Romantic notions of AI existing in a cloud are far from reality. The technology depends on energy-hungry data centres, around half of which are in the US.

The Action Plan calls for streamlined permitting, underlined by an executive order also announced yesterday; there's no point in Alphabet's Google pledging to make \$85bn in capital expenditure this year, as it did yesterday, if it can't get the permits to build the data centres or the power to run them. The plan also makes the case for a revitalised domestic semiconductor-manufacturing sector; and greater security both for corporate and military and intelligence users.

Much of the focus though is on energy. Power usage by data centres in the US more than doubled from less than 2 percent in 2018 to 4.4 percent in 2023, and is set to rise to as much as 12 percent, or 580 TWh in 2028, according to the US Department of Energy's Lawrence Berkeley National Laboratory.

Figure 2: Total US data centre electricity use from 2014 through 2028 (TWh)



Source : Lawrence Berkeley National Laboratory, Deutsche Bank

One leading startup, Anthropic, said in a recent [white paper](#) that the US AI sector will need at least 50GW of electric capacity by 2028, with 5GW data centres needed to train the most advanced models.

Other countries, such as Ireland, have seen data centres consuming more than 20 percent of the country's electricity; the US does not want to follow. After two decades of stagnant demand, strain is being put on the system not just by growth in data centres but even more so by a surge in industrial use and electric transport.

There has already been some progress. Investments in the electricity grid have grown worldwide after generally falling in the late 2010s, but since 2020 they have taken off again – and nowhere more so than in North America.



A total of \$114bn was invested in the grid there last year, up from the mid-\$70bns each year in the latter half of the 2010s, according to the International Energy Agency. China invested \$89bn in its grid last year and Europe \$85bn, the data show.

Now the US Action Plan foresees preventing the premature decommissioning of critical power generation resources; upgrading existing transmission systems and power lines to increase the amount of energy that can be transmitted; and encouraging the expansion of the grid with new technologies such as enhanced geothermal, nuclear fission and nuclear fusion.

3. International AI: projecting power

Just because the US is ahead of rivals today does not mean it will be tomorrow.

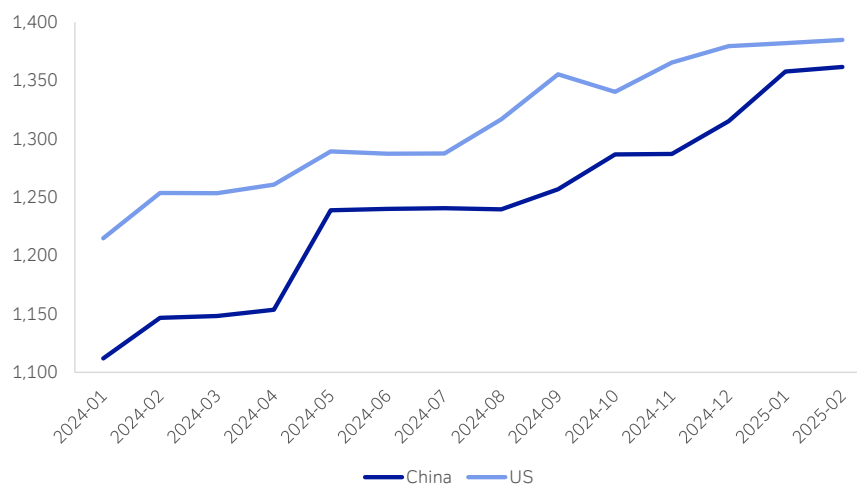
The final portion of the Action Plan focuses on national security – emphasised by the fact that the third signatory to the Action Plan was Secretary of State Marco Rubio. It stresses that the US should lead an “enduring global alliance, while preventing our adversaries from free-riding on our innovation and investment”.

The launch of a cutting-edge AI model by China’s DeepSeek in January caused a rout in US tech stocks, including an 18 percent drop in the market cap of Nvidia. The initial focus was on how the model had apparently been developed for a fraction of the cost of comparable US models, potentially calling into question several hundred billion dollars of investment in the most advanced AI chips at home.

But the more troubling aspect from a US national security point of view was that either the tight controls on exports of those chips and other AI components to China were not effective – or that they were effective – but that Chinese inventiveness with older chips or home-grown equivalents meant it didn’t matter.

As the chart shows, Chinese models are rapidly approaching the capabilities of those from the US. The performance gap on major benchmarks is closing, with the top US models scoring 1,385 in LMSYS Chatbot Arena as of February, barely ahead of the 1,362 scored by their Chinese counterparts.

Figure 3: Chinese models are catching up with US on Chatbot Arena



Source : Stanford University HAI AI Index Report 2025, Deutsche Bank



China also dominates in AI patents, accounting for 70 percent of grants in 2023, trouncing the US with just 14 percent, according to the Stanford AI Index. Whatever the respective quality of those patents, US policymakers clearly do not want to appear complacent.

The Action Plan looks to put the weight of the Department of Commerce towards selling AI packages to allies, codified in another new executive order on "[promoting the export](#) of the American AI technology stack". It also pledges to invest in areas such as biosecurity.

However, most of the emphasis is on throwing obstacles in the way of China's use of AI to gain a competitive edge.

That includes using location verification features on advanced AI compute to tighten export control enforcement; developing new controls to plug loopholes, such as exports of component sub-systems; and compelling allies to ensure they reinforce export controls and imposing penalties such as tariffs on them if they do not.

It remains to be seen how this will work in practice in the face of Realpolitik: last week the administration said it would relax export controls on Nvidia and AMD selling some (less sophisticated) AI chips to China in return for China restarting its exports of some rare earths to the US.

[Some underlying reassurance: jobs, ideology and safety](#)

The Action Plan also lays out some overarching principles in an attempt to reassure a sceptical public. A global Ipsos poll last year showed that Americans are among the least enthusiastic citizens worldwide about AI, with fewer than 40 percent agreeing that it has more benefits than drawbacks.

The plan tells American workers that AI will enhance their lives by complementing their work, not replacing it. AI systems must be trustworthy and "free from ideological bias", it says – accompanied by a third new executive order called "[preventing woke](#) AI in the federal government" to prevent agency heads from procuring AI models that fail to meet its standards for truth and neutrality. And there will be "constant vigilance" of misuse, theft or "emerging and unforeseen risks".



Appendix 1

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